

MARKOV CHAINS

Summary — Graduate Level

Definitions · Classification · Asymptotic Behavior · Limit Theorems

Markov Property

Irreducibility

Recurrence

Ergodicity

DEFINITIONS & TRANSITION MATRIX

Homogeneous Markov Property

$$P(X_{n+1} = j \mid X_0 = i_0, \dots, X_n = i) = p_{ij}$$

- $P = (p_{ij})_{i,j \in E}$ stochastic: $p_{ij} \geq 0$, $\sum_j p_{ij} = 1$
- Distribution at time n : $\mu_n = \mu_0 P^n$

$$\text{Chapman-Kolmogorov: } P^{m+n} = P^m P^n, \text{ i.e. } p_{ij}^{m+n} = \sum_k p_{ik}^m p_{kj}^n$$

Recurrence & Transience

$$T_i = \inf\{n \geq 1 : X_n = i\}, \quad f_{ii} = P_i(T_i < \infty)$$

- i recurrent iff $f_{ii} = 1$ iff $\sum_n p_{ii}^n = \infty$
- i transient iff $f_{ii} < 1$ iff $\sum_n p_{ii}^n < \infty$

$$m_i = E_i[T_i]: \text{ positive rec. iff } m_i < \infty, \text{ null rec. iff } m_i = \infty$$

In a finite closed class: all states are positive recurrent

Communication Classes

$$i \rightarrow j \text{ iff exists } n \geq 0 : p_{ij}^{(n)} > 0$$

- $i \leftrightarrow j$ (equivalence) if $i \rightarrow j$ and $j \rightarrow i$
- Irreducible chain: E is a single communication class

$$\text{Period: } d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}$$

$d(i)$ is constant within each communication class

Stopping Times & Strong Markov

$$T \text{ stopping time: } \{T = n\} \text{ in } \sigma(X_0, \dots, X_n) \text{ for all } n$$

Strong Markov property: conditionally on $\{T < \infty, X_T = i\}$,

$$(X_{T+n})_{n \geq 0} \text{ is a MC}(p_{ij}) \text{ independent of } \mathcal{F}_T$$

- Number of visits to i : $N_i = \sum_{n \geq 1} 1_{\{X_n = i\}}$

$$E_i[N_i] = f_{ii} / (1 - f_{ii}) \text{ if transient, } = \infty \text{ if recurrent}$$

STATIONARY MEASURE & CONVERGENCE

1 Stationary Measure

π stationary iff $\pi P = \pi$, $\sum_i \pi(i) = 1$

- Irred. + pos. rec. $\Rightarrow$ exists! π , with $\pi(i) = 1/m_i$
- If $\mu_0 = \pi$ then $\mu_n = \pi$ for all n (stationarity)

Detailed balance condition (reversibility):

$\pi(i) p_{ij} = \pi(j) p_{ji}$ for all $i, j \Rightarrow \pi$ stationary

3 Convergence in Total Variation

If irred. + pos. rec. + aperiodic:

$$\|P^n(i, \cdot) - \pi\|_{TV} = (1/2) \sum_j |p_{ij}^{(n)} - \pi(j)| \rightarrow 0$$

- Rate: $\rho = \max\{|\lambda| : \lambda \text{ in Spec}(P), \lambda \neq 1\} < 1$
- Exponential speed: $\|P^n - \pi\| \leq C \rho^n$

Mixing time: $T_{\text{mix}}(\epsilon) = \inf\{t : \max_i \|P^t(i, \cdot) - \pi\|_{TV} \leq \epsilon\}$

2 Ergodic Theorem (Birkhoff)

If irreducible + positive recurrent, for all f in $L^1(\pi)$:

$$(1/n) \sum_{k=0}^{n-1} f(X_k) \rightarrow \pi(f) = \sum_i \pi(i) f(i) \quad P\text{-a.s.}$$

- For $f = 1_{\{j\}}$: $(1/n) \#\{k < n : X_k = j\} \rightarrow \pi(j)$ a.s.
- Holds regardless of the initial distribution

for all i : $P_i(X_n = j \text{ i.o.}) = 1$ if j is positive recurrent

4 Central Limit Theorem for Markov Chains

If irred. + pos. rec. + aperiodic, f in $L^2(\pi)$:

$$(1/\sqrt{n}) \sum_{k=1}^n [f(X_k) - \pi(f)] \rightarrow N(0, \sigma^2(f))$$

$$\sigma^2(f) = \text{Var}_{\pi}(f) + 2 \sum_{n=1}^{\infty} \text{Cov}_{\pi}(f(X_0), f(X_n))$$

$$= \langle f - \pi(f), (I + P + P^2 + \dots)(f - \pi(f)) \rangle_{\pi}$$

RECURRENCE CRITERIA & CLASSICAL EXAMPLES

Foster-Lyapunov Criterion

Exists $V: E \rightarrow [1, \infty)$, C subset E finite, $\epsilon > 0$, $b < \infty$:

$$PV(i) = \sum_j p_{ij} V(j) \leq V(i) - \epsilon + b \cdot 1_C(i)$$

$\Rightarrow$ positive recurrence + $E_i[T_i] \leq V(i)/\epsilon + b/\epsilon$

$PV \leq \lambda V + b \cdot 1_C$ with $\lambda < 1 \Rightarrow$ geometric return time

Case $\lambda < 1$: exponential convergence is guaranteed

Random Walk on $\mathbb{Z}^d$ (Polya's Theorem)

$$p(x, x + e_k) = 1/(2d) \text{ for } k = 1, \dots, d$$

$$p_{xx}^{(2n)} = (1/(2\pi)^d) \int_{[-\pi, \pi]^d} (\cos \theta_1 + \dots + \cos \theta_d)^{2n} / d^d d\theta$$

- $d = 1, 2$: recurrent ($\sum p_{xx}^{(2n)} = \infty$) [Polya 1921]
- $d \geq 3$: transient ($\sum p_{xx}^{(2n)} < \infty$)

Birth-Death Chain

$$p_{i,i+1} = \lambda_i, p_{i,i-1} = \mu_i, p_{ii} = 1 - \lambda_i - \mu_i$$

Stationary measure (if it exists):

$$\pi(n) = \pi(0) \cdot (\lambda_0 \lambda_1 \dots \lambda_{n-1}) / (\mu_1 \mu_2 \dots \mu_n)$$

Positive rec. iff $\sum_n \pi(n) < \infty$ (normalizable)

Pure criterion: $\sum_n (\mu_1 \dots \mu_n) / (\lambda_1 \dots \lambda_n) = \infty$ iff recurrent

M/M/1 Queue

$$\lambda_i = \lambda, \mu_i = \mu \ (i \geq 1), \rho = \lambda/\mu$$

$$\rho < 1 \Rightarrow \pi(n) = (1 - \rho) \rho^n, n \geq 0$$

$$E_{\pi}[N] = \rho/(1 - \rho), \text{Var}_{\pi}[N] = \rho/(1 - \rho)^2$$

$\rho \geq 1 \Rightarrow$ transient ($\rho > 1$) or null recurrent ($\rho = 1$)

SPECTRUM, REVERSIBILITY & ALGORITHMS

Spectral Analysis (finite state space)

$$\text{Spec}(P) = \{\lambda_1 = 1 \geq \lambda_2 \geq \dots \geq \lambda_N \geq -1\}$$

If P reversible (self-adjoint in $L^2(\pi)$): real eigenvalues

$$\text{Spectral gap: } \gamma^* = 1 - \max(\lambda_2, |\lambda_N|) > 0 \text{ iff ergodic}$$

$$\|P^n(i, \cdot) - \pi\|_{TV} \leq (1/\sqrt{\pi(i)}) (1 - \gamma^*)^n$$

$$T_{\text{mix}}(\epsilon) \leq (1/\gamma^*)(\log(1/\pi_{\min}) + \log(1/\epsilon))$$

Poincare & Cheeger Inequalities

Poincare constant $\lambda_2 = 1 - \text{gap}(P)$:

$$\text{Var}_\pi(f) \leq (1/\gamma) E(f, f), \quad E(f, f) = (1/2) \sum_{i,j} \pi(i) p_{ij} (f(j) - f(i))^2$$

$$\text{Cheeger constant: } h = \min_{S: \pi(S) \leq 1/2} Q(S, S^c) / \pi(S)$$

$$h^2/2 \leq \gamma^* \leq 2h \text{ (Cheeger inequalities)}$$

Metropolis-Hastings Algorithm

Proposal kernel $Q(i, j)$, target π . Acceptance rate:

$$\alpha(i, j) = \min(1, \pi(j) Q(j, i) / \pi(i) Q(i, j))$$

$$\text{MH kernel: } P(i, j) = Q(i, j) \alpha(i, j) \text{ for } j \neq i$$

$P(i, i) = 1 - \sum_{j \neq i} P(i, j)$ (rejection probability)

Detailed balance: $\pi(i)P(i, j) = \pi(j)P(j, i) \Rightarrow \pi$ stationary

Coupling & Total Variation Distance

$$\|\mu - \nu\|_{TV} = \max_{A \subset E} |\mu(A) - \nu(A)| = (1/2) \sum_i |\mu(i) - \nu(i)|$$

$$\|\mu - \nu\|_{TV} = \inf \{ P(X \neq Y) : (X, Y) \text{ coupling of } (\mu, \nu) \}$$

Coupling time: $\tau_{\text{coup}} = \inf \{ n : X_n = Y_n \}$

$$\|P^n(i, \cdot) - \pi\|_{TV} \leq P_{\{i, \pi\}}(\tau_{\text{coup}} > n)$$

Markov Chains — Concept Map

Markov Property

$$P(X_{n+1}=j|F_n) = p_{ij}$$

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Irreducibility

$$\text{for all } i, j: p_{ij}^{(n)} > 0$$

+

+

Positive Recurrence

$$m_i = E_i[T_i] < \infty$$

Aperiodicity

$$d(i) = \gcd\{n: p_{ii}^{(n)} > 0\} = 1$$

Markov Chain

Invariant Measure

$$\pi P = \pi, \pi(i) = 1/m_i$$

Ergodic Theorem

$$(1/n) \sum f(X_k) \rightarrow \pi(f) \text{ a.s.}$$

TV Convergence

$$\|P^n(i, \cdot) - \pi\|_{TV} \rightarrow 0$$

Irreducible + Positive Recurrent + Aperiodic => ERGODIC Chain